

DBSCAN: From First Principles

DBSCAN is a clustering algorithm that groups points based on density. It can find clusters of unusual shapes and can identify noise points.

Why DBSCAN Is Needed

K-Means assumes clusters are roughly round and requires choosing K. DBSCAN does not require K and can handle clusters with irregular shapes.

Core Terms

The two main hyperparameters are ϵ and `min_samples`. ϵ defines the neighborhood radius. `min_samples` defines how many points are needed for a dense region.

Core Points, Border Points, and Noise

A core point has enough neighbors within ϵ . A border point is near a core point but does not have enough neighbors itself. A noise point does not belong to any dense region.

How It Clusters

DBSCAN starts from core points and expands clusters by connecting nearby dense regions. Points not connected to dense regions become noise.

When to Use It

Use DBSCAN when clusters have irregular shapes or when noise detection matters. Avoid it when density varies strongly across clusters.